MIL-G-0015342C(SHIPS)
8 October 1975
USED IN LIEU OF
MIL-G-15342B
16 August 1971
(See 6.4)

MILITARY SPECIFICATION

GASKETS, METALLIC ASBESTOS, SPIRAL WOUND, FOR BOILERS

This limited coordination Military Specification has been prepared by the Naval Ship Engineering Center based upon currently available technical information and is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense. It has not been approved for promulgation as a coordinated revision of Military Specification MIL-G-15342B. It is subject to modification. However, pending its promulgation as a coordinated Military Specification, it may be used in procurement.

1.1 Scope. This specification covers metallic-asbestos gaskets for handholes and manholes of shipboard and stationary boilers. Nuclear boilers are excluded.

1.2 Classification. Gaskets shall be of the following classes and series, as specified (see 6.2):

Class 1 - Boiler handhole.
Class 2 - Boiler manhole.

Series 600
Series 1500

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
QQ-S-766 - Steel Plates, Sheets, and Strip - Corrosion Resisting.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-576 - Box, Wood, Cleated, Veneer, Paper Overlaid.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Shipping, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-B-665 - Boxes; Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676 - Boxes, Set-up.

MILITARY
MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.
MIL-A-24058 - Asbestos Packing and Gasketing Material, Impregnated.

STANDARDS

FEDERAL
FED-STD-151 - Metals; Test Methods.

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

MILITARY
NAVSHIPS Dwg 803-841218 - Gasket, Boiler Handhole and Manhole, 1500 PSI Maximum Pressure and 950°F Maximum Temperature.

FSC 5330

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:05Z
Check the source to verify that this is the current version before use.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION INCORPORATED, AGENT
National Motor Freight Classification Rules.

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., 1616 P Street, N.W., Washington, D.C. 20036.)

AMERICAN IRON AND STEEL INSTITUTE (AISI)

(Application for copies should be addressed to the American Iron and Steel Institute, 150 East 42nd Street, New York, N.Y. 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
A 167-74 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip, Spec. for.
D 1161-60 - Total Chloride Content of Paper and Paper Products, Test for.
D 1918-72 - Asbestos Content of Asbestos Textile Materials, Test for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Qualification. Metallic-asbestos gaskets furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.3).

3.2 Construction. Gaskets shall be composed of a single thickness of metal wound in a spiral shape, starting on the inside and working outward ply by ply to the desired size. Between the plies of metal shall be a cushion of asbestos filler material wound to conform with the metal strip. For class 1 and 2 gaskets, spot welds, shall be as shown on NAVSHIPS dwg 803-841218.

3.2.1 Dimensions and tolerances. Unless otherwise specified (see 6.2), gaskets shall conform to the dimensions, tolerances, and other data shown, as applicable to this specification, on the drawing. Class 2 gaskets shall be a true ellipse.

3.2.2 Plies. A ply shall consist of one 360-degree turn of the metal or asbestos strip. Gaskets shall be wound on a mandrel of the desired shape. After one complete revolution, the initial spot welds shall be made. Approximately midway during the next revolution, the asbestos filler strip shall be introduced. Winding shall continue to make the furnished product contain the number of plies of asbestos required. A minimum of two plies of metal shall then be made and then the terminal spot welds.

3.3 Materials.

3.3.1 Metal strip. The metal strip used in the gasket shall be one of the following materials:

(a) Corrosion resisting steel, AISI type 347, cold rolled and annealed, in accordance with ASTM A 167-74 or QQ-S-766, type 347, cold rolled and annealed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:05Z
Check the source to verify that this is the current version before use.

(b) Corrosion resisting steel, AISI type 316L, in accordance with ASTM A 167-74, or QQ-S-766, type 316L.

(c) Corrosion resisting steel, AISI type 309S, in accordance with the applicable requirements of ASTM A 167-74 except that type 309S shall be stabilized with columbium (and tantalum not exceeding 0.10 percent) and their combination to be not less than 10 times the carbon content; or QQ-S-766, type 309S, stabilized and combined as noted above.

3.3.1.1 Thickness. The thickness of the metal stripshall be between 0.0063 and 0.0088 inch.

3.3.2 Asbestos filler. The asbestos paper filler shall consist of chrysotile asbestos fiber conforming to type III of MIL-A-24058, but with not more than 10 percent binder content of suitable rubber or paraffin.

3.4 Service conditions.

3.4.1 Service conditions and temperature ratings. Gaskets shall be rated as suitable for service at the following temperatures:

Class 1 - Handhole plate gaskets:
Series 600 - 850° Fahrenheit (°F)
Series 1500 - 950°F
Class 2 - Manhole plate gaskets:
Series 600 - 500°F
Series 1500 - 600°F

3.4.2 Service condition of bearing surface. Gaskets shall be suitable for service in connection with flat smooth surfaces having finishes of spiral or concentric lay of 60 to 80 serrations per inch and a roughness height rating of 125.

3.5 Marking. Gasket shall be marked with a sturdy paper tag (of tag stock) securely attached thereto, which shall contain thereon the following information (see 6.2): The manufacturer's name; gasket size, including shape, thickness, inside and outside dimensions; type of metal and filler material; pressure and temperature rating, National stock number, and customer's order number. For class 1, at least one tag shall be applied to each 25 gaskets or less. For class 2 gaskets, each gasket shall be tagged.

3.6 Workmanship. The finished gaskets shall have no breaks, creases, or sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Qualification inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

4.3 Qualification tests. Qualification tests shall be conducted at a laboratory satisfactory to the Naval Ship Engineering Center. Qualification tests shall consist of the examinations of tests specified in 4.5. Manufacturers shall furnish a minimum of six representative samples for each class and series of gasket for which qualification is desired.

4.4 Quality conformance inspection. Each of the sample gaskets selected in accordance with 4.4.1.3 shall be analyzed in accordance with method 111 of FED-STD-151 to determine compliance with 3.2.1.1. If any sample fails to conform to 3.2.1.1, this shall be cause for rejection of the entire lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:05Z
Check the source to verify that this is the current version before use.

4.4.1 Sampling.

4.4.1.1 Lot. All gaskets of the same size and series manufactured from the same batch of material and presented at one time shall be considered a lot for purposes of examination and tests.

4.4.1.2 Sampling for examination. A random sample of gaskets shall be selected from each lot of material offered for inspection of visual and dimensional characteristics in accordance with the sampling procedures of MIL-STD-105, Special Inspection Level S-1. Acceptable Quality Level (AQL) shall be 1.5 percent defective.

4.4.1.3 Sampling for chemical analysis.

4.4.1.3.1 Metal strip. Five feet of uncrimped metal strip representing the lot shall be forwarded to the testing facility for purposes of chemical analysis of metal.

4.4.1.3.2 Asbestos. Twenty grams of asbestos paper representing the lot shall be forwarded to the testing facility for purposes of chemical analysis of asbestos.

4.4.2 Visual and dimensional examination. Each of the sample gaskets selected in accordance with 4.4.1.2 shall be visually and dimensionally examined to verify compliance with this specification. Any gasket in the sample containing one or more defects shall not be offered for delivery, and if the number defective exceeds the acceptance number, this shall be cause for rejection of the lot represented by the sample. Examination shall be conducted as specified in table I.

Table I - Classification of defects.

Categories	Defects
Critical	None defined.
Major:	
101	Class of gasket not as specified.
102	Rating of gasket not as specified (pressure series).
103	Thickness of finished gasket not within specified tolerances.
104	Dimensioning of gasket not as specified in contract or on standard drawing.
105	Unnecessary breaks, creases, or sharp edges on finished gaskets.
106	Identification markings missing on tags, illegible, incorrect, or not as specified.
107	Packaging or packing not as specified.
108	Number of metal plies less than specified minimum.
109	Number of asbestos plies less than minimum specified, or plies less than a 360 degrees each where specified.
110	Spot welds missing, not minimum number, incorrectly spaced, or defective.
111	Thickness of metal strip not within specified minimum-maximum, or not continuous.
Minor	None defined.

4.5 Test procedures.

4.5.1 Chemical analysis of metal strip. The metal strip selected in compliance with 4.4.1.3.1 shall be analyzed in accordance with method 111 of FED-STD-151 to determine compliance with 3.2.1.1. If any sample fails to conform, this shall be cause for rejection of the entire lot.

4.5.2 Chemical analysis of asbestos. The asbestos paper selected in compliance with 4.4.1.3.2 shall be analyzed for conformance with 3.2.2.1. Test for leachable chloride shall be in accordance with MIL-A-24058. Test for total chloride shall be in accordance with ASTM D 1161-60. Test for asbestos content shall be in accordance with ASTM D 1918-72. If any sample fails to conform, this shall be cause for rejection of the entire lot.

4.6 Possible test failures. Possible test failures are defined as follows:

(a) Steel strip not as specified.
(b) Asbestos filler not as specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:05Z
Check the source to verify that this is the current version before use.

4.7 Comparison inspection. Comparison inspection shall be conducted to verify continuing quality of the approved product and compliance with the specification requirements. These tests will be on samples representing those series and sizes which were tested for qualification and approved, and will be conducted on six samples of each series and size, supplied by the manufacturer at no cost to the Government. Comparison inspection is not required more often than every two years unless the inspector has reason to consider it necessary. Comparison examination and tests shall be those specified in 4.4.2 and 4.5 plus any others specifically requested by the inspector.

4.8 Inspection of preparation for delivery. Sample packages and packs and the inspection of the packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements.)

5.1 Packaging. Packaging shall be level A or level C, as specified (see 6.2).

5.1.1 Level A. Gaskets, quantity as specified (see 6.2), shall be packaged in boxes conforming to PPP-B-566, PPP-B-636, PPP-B-665, or PPP-B-676, at the option of the supplier. Box closure shall be as specified in the applicable box specification or appendix thereto.

5.1.1.1 Bundling. Gaskets shall be bundled in quantities, tied, and wrapped in a non-corrosive, dust protective opaque or transparent paper or film as follows:

Class 1 - 25 gaskets per bundle
Class 2 - 12 gaskets per bundle

The wrapper shall be secured with pressure sensitive tape or heat sealed.

5.1.2 Level C. Gaskets shall be packaged in a manner to afford protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and immediate use. The supplier's normal packaging method may be used when such meets the requirements of this level.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Gaskets, packaged as specified (see 6.2), shall be packed in fiberboard boxes conforming to class weather-resistant of PPP-B-636. Alternately, gaskets may be packed in containers conforming to any one of the following specifications at the option of the supplier:

Specification	Type or class
PPP-B-576	Class 2
PPP-B-585	Class 3
PPP-B-591	Class II
PPP-B-601	Overseas type
PPP-B-621	Class 2
PPP-B-640	Class 2
PPP-B-665	Weather Resistant

Shipping containers shall have caseliners conforming to MIL-L-10547. Caseliners shall be closed and sealed in accordance with the appendix to MIL-L-10547. Boxes shall be closed, strapped or banded in accordance with the applicable box specification or appendix thereto, except that PPP-B-636 boxes shall be closed method V in accordance with the appendix to the box specification and caseliners are not required. The gross weight of wood or wood-cleated boxes shall not exceed 200 pounds.

5.2.2 Level B. Gaskets, packaged as specified (see 6.1), shall be packed in fiberboard boxes having a minimum bursting strength of 275 lb/in^2, conforming to PPP-B-636, class domestic. Alternatively, gaskets may be packed in containers conforming to any one of the following specifications at the option of the supplier:

Specification	Type or class
PPP-B-576	Class 1
PPP-B-585	Class 1 or 2 use
PPP-B-591	Class I

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:05Z
Check the source to verify that this is the current version before use.

Specification	Type or class
PPP-B-601	Domestic type
PPP-B-621	Class 1
PPP-B-640	Class 1
PPP-B-665	Weather resistant

5.2.2.1 Box closure shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood or wood-cleated boxes shall not exceed 200 pounds. Not more than one size or type of gaskets shall be packed in a single container.

5.2.3 Level C. Gaskets of like series and size, packaged as specified (see 6.2). shall be packed in containers which will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply to the Uniform Freight or National Motor Freight Classification Ratings, Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

5.2.4 Filling of voids. Voids or cavities in the shipping containers shall be filled with cushioning materials to prevent shifting of contents during handling, shipment and storage. The use of loose excelsior, shredded newspaper and similar materials which have not been refined for non-corrosiveness shall not be used unless these materials were enclosed in a sealed waterproof barrier material. Use of expanded, loose-fill, polystyrene is permitted when approved (see 6.2) by the procuring activity.

5.3 Marking. In addition to any special marking required by the contract or order (see 6.2), interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The metallic-asbestos gaskets covered by this specification are intended for use in handholes and manholes of shipboard and stationary boilers with temperatures not exceeding 950°F and 1500 lb/in^2 maximum pressure.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, mumber, and date of this specification.
(b) Class and series of gasket (see 1.2).
(c) Line item number for gasket from standard drawing, or if not listed, a full description such as where used in boiler, inside and outside dimensions, thickness, bearing surface and all other pertinent information (see 3.2.1 and 3.5).
(d) Level of packaging and packing (see 5.1 and 5.2).
(e) Quantity per unit package (see 5.1.1).
(f) When use of expanded, loose-fill polystyrene is approved (see 5.2.4).
(g) Special marking, if required (see 5.3).

6.3 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in applicable Qualified Products List QPL-0015342 whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Ship Engineering Center, Prince George's Center, Center Building, Hyattsville, Maryland 20872, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3.1).

6.3.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.4 THE MARGINS OF THIS SPECIFICATION ARE MARKED "#" TO INDICATE WHERE CHANGES (ADDITIONS, MODIFICATIONS, CORRECTIONS, DELETIONS) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS WAS DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

Preparing activity:
Navy - SH
(Project 5330-N043)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:05Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

☆ U.S. GOVERNMENT PRINTING OFFICE: 1972 — 714-862/3404-2:1

Source: https://assist.dla.mil - Downloaded: 2026-10-19T06:05Z
Check the source to verify that this is the current version before use.

FOLD

COMMANDER
NAVAL SHIP ENGINEERING CENTER
CENTER BUILDING - SEC 6124
PRINCE GEORGES CENTER
HYATTSVILLE, MARYLAND 20782

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

POSTAGE AND FEES PAID
DEPARTMENT OF NAVY

DOD 316

U.S.MAIL

COMMANDER
NAVAL SHIP ENGINEERING CENTER
CENTER BUILDING - SEC 6124
PRINCE GEORGES CENTER
HYATTSVILLE, MARYLAND 20782

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:05Z
Check the source to verify that this is the current version before use.